

18 B connecting wire so all metallic surfaces at same potential

$$V_1 = V_2$$

$$\left(\frac{1}{4\pi\epsilon_0} \right) \frac{q_1}{r_1} = \left(\frac{1}{4\pi\epsilon_0} \right) \frac{q_2}{r_2}$$

$$\frac{E_1}{E_2} = \frac{\left(\frac{1}{4\pi\epsilon_0} \right) \left(\frac{q_1}{r_1^2} \right)}{\left(\frac{1}{4\pi\epsilon_0} \right) \left(\frac{q_2}{r_2^2} \right)} = \frac{\left(\frac{q_1}{r_1} \right) \frac{1}{r_1}}{\left(\frac{q_2}{r_2} \right) \frac{1}{r_2}} = \frac{r_2}{r_1}$$

7

19 **D** electric force is vector quantity and the charge at X is negative so electric field strength actually